# Spencer H. Bryngelson

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# 1 Education

- University of Illinois at Urbana–Champaign
  - (2017) Doctor of Philosophy, Theoretical & Applied Mechanics
    Advisor: Jonathan Freund
  - (2015) Master of Science, Theoretical & Applied Mechanics
  - (2015) Graduate Certificate, Computational Science & Engineering
- University of Michigan–Dearborn
  - (2013) Bachelor of Science, Mechanical Engineering
  - (2013) Bachelor of Science, Engineering Mathematics

# 2 Positions held

- (2026–Present) Associate Professor, Walker Department of Mechanical Engineering, Cockrell School of Engineering, The University of Texas at Austin
- (2026–Present) Walker Scholar, Walker Department of Mechanical Engineering, The University of Texas at Austin
- (2026–Present) Core Faculty, Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin
- (2021–26) Assistant Professor, School of Computational Science & Engineering, College of Computing, Georgia Institute of Technology
- (2023–26) Assistant Professor by Courtesy (0%), Daniel Guggenheim School of Aerospace Engineering, College of Engineering, Georgia Institute of Technology
- (2024–26) Assistant Professor by Courtesy (0%), George W. Woodruff School of Mechanical Engineering, College of Engineering, Georgia Institute of Technology
- (2025–26) Associate Director, Center for Research into Novel Compute Hierarchies (CRNCH), Georgia Institute of Technology
- (2025) Visiting Professor, Pacific Northwest National Laboratory, Department of Energy
- (2025) Summer Faculty Research Program, Office of Naval Research, NSWC Carderock
- (2022) Visiting Scholar, Stanford University, Center for Turbulence Research (Summer Program)
- (2018–21) Senior Postdoctoral Scholar, California Institute of Technology, with Tim Colonius
- (2019) Visiting Researcher, Massachusetts Institute of Technology, with Themis Sapsis
- (2018) Postdoctoral Researcher, XPACC (PSAAP II center), with Carlos Pantano, Dan Bodony, Jon Freund
- (2013–17) Graduate Research Fellow, University of Illinois at Urbana–Champaign, with Jon Freund
- (2015) Alumni Teaching Fellow, University of Illinois at Urbana–Champaign
- (2012–13) Undergraduate Research Assistant, University of Michigan–Dearborn, with Eric Ratts

# 3 Teaching

## 3.1 Georgia Institute of Technology

| Semester | Number | Course Title | Students |
|---|---|---|---|
| Spring 2026 | CSE 6730 | Modeling & Simulation | 183 |
| Fall 2025 | CSE 6010 | Computational Problem Solving | 39 |
| Spring 2025 | CSE 6730 | Modeling & Simulation | 180 |
| Fall 2024 | CX/MATH 4640 | Numerical Analysis I | 45 |
| Spring 2024 | CSE 6730 | Modeling & Simulation | 163 |

| | | | |
|---|---|---|---|
| Fall 2023 | CX/MATH 4640 | Numerical Analysis I | 53 |
| Spring 2023 | CSE 6730 | Modeling & Simulation | 146 |
| Fall 2022 | CX/MATH 4640 | Numerical Analysis I | 36 |
| Fall 2021 | CX/MATH 4640 | Numerical Analysis I | 43 |

**Note:** I co-teach VIP (2/3/3/4)60(1/2) *Team Phoenix: Cluster Competition Team (HPC)* with Prof. R. Vuduc each Fall and Spring since Fall 2022.

### 3.2 Other institutions

| Semester | Number | Course Title | Students | Institute |
|---|---|---|---|---|
| Fall 2015 | ME310 | Fundamentals of Fluid Dynamics | 82 | UIllinois |
| Fall 2013 | ME3601 | Design and Analysis of Machine Elements | 35 | UMichigan |
| Spring 2012 | ME364 | Probability, Statistics, and Reliability in Design | 32 | UMichigan |
| Fall 2012 | ME230 | Statics and Mechanics of Materials | 61 | UMichigan |

## 4 Students

### 4.1 Staff

- Dr. Tianyi Chu, Postdoctoral Fellow

### 4.2 Ph.D.

- Jesus Arias (CSE/AE, Georgia Tech)
- Daniel Vickers (CSE, Georgia Tech)
- Andrew Wang (CSEM, UT Austin)
- Benjamin Wilfong (CSE, Georgia Tech)

### 4.3 Undergraduate

- Melody Lee (CS, Georgia Tech)

### 4.4 Alumni

#### 4.4.1 Graduate students

- Anand Radhakrishnan, Ph.D. CSE (Georgia Tech), 2021–26. Next: NVIDIA
- Haocheng Yu, Ph.D. CSE (Georgia Tech), co-advised with K. Ahuja, 2021–26
- Zhixin (Jack) Song, Ph.D. Physics (Georgia Tech), 2021–26. Next: Quantum Research Scientist, QuEra
- Fatima Ezahra Chrit, Ph.D. ME (Georgia Tech), co-advised with Alex Alexeev, 2021–23. Next: CFD Engineer, IronCAD
- Anshuman Sinha, M.S. CSE (Georgia Tech), 2022–23. Next: Intern, LLNL; ML engineer, Apple

#### 4.4.2 Undergraduate students

- Mohammed Al-Mahrouqi, AE (Georgia Tech), 2025
- Tanush Prathi, CS (Georgia Tech), 2025
- Ajay Bati, CS (Georgia Tech), 2021–23. Next: Intern, Apple
- Arjun Bhamra, CS (Georgia Tech), 2022–23. Next: Intern, Quantinuum; Intern, IBM; Researcher, TINKER Lab (T. Conte, GT)
- Rasmit Devkota, Physics (Georgia Tech), 2023. Next: Research Scientist (GTRI)
- Ansh Gupta, CS (Georgia Tech), 2022–24. Next: YCombinator startup, co-founder

- Yash Kothari, CS (Georgia Tech), 2022–23. Next: Co-founder, Inika AI
- Henry Le Berre, CS (Georgia Tech), 2021–25. Next: Propulsion Software Engineer, SpaceX
- Suzan Manasreh, CS (Georgia Tech), 2024. Next: Intern, NVIDIA; BSMS CS (GT)
- Sriharsha Kocherla, CS (Georgia Tech), 2022-24. Next: Intern (Quant, Chicago Trading); MSCS (GT); Quant., Old Mission
- Subrahmanyam Mullangi, CS (Georgia Tech), 2023–24. Next: Xantium Group, Tudor
- Brian Ok, CS (Georgia Tech), 2024–25 (GT OSPO VSIP Program).
- Lian Xiang (Physics, Georgia Tech). Next: Ph.D. student, RPI
- Qi Zeng, CS & Math (Georgia Tech), co-advised with F. Schäfer, 2021–23. Next: MSCS (UCBerkeley); ML Engineer, TikTok; Software Engineer, Google Cloud AI

### 4.5 Student and scientist accolades

- (2026) Melody Lee, Barry Goldwater Scholar
- (2025) Melody Lee, GT PURA Salary Award
- (2025) Benjamin Wilfong, Winner, OLCF User Group Visualization Showcase
- (2024) Benjamin Wilfong, GT CRNCH Fellowship
- (2024) Elizabeth Hong, GT PURA Salary Award
- (2024) Suzan Manasreh, GT PURA Salary Award
- (2024) Subrahmanyam Mullangi, GT PURA Salary Award
- (2023) Dr. Bryan Gard (GTRI Research Scientist), IRAD of the Year award
- (2023) Qi Zeng, Outstanding Undergraduate Researcher Award, College of Computing (co-advised with F. Schäfer)
- (2023) Ansh Gupta, GT PURA Salary Award
- (2022) Fatima Chrit, Georgia Tech Quantum Alliance Fellowship
- (2022) Zhixin (Jack) Song, GT CRNCH Fellowship
- (2022) Benjamin Wilfong, GT President's Fellowship

## 5 Awards

- (2026) NATO Science & Technology Organization (STO) Excellence Award, Research Task Group AVT-346, "*Predicting Hypersonic Boundary-Layer Transition on Complex Geometries*"
- (2025) ACM Gordon Bell Prize, Finalist
- (2025) Winner, OLCF User Group Visualization Showcase
- (2024) Scaling Efficiency Certificate, Jülich Supercomputing Centre JUPITER Research and Early Access Program (JUREAP)
- (2022) Ralph E. Powe Junior Faculty Enhancement Award, Oak Ridge National Lab
- (2022–23) Georgia Tech Faculty Writing Scholar
- (2022–23) Class of 1969 Teaching Fellow, Georgia Institute of Technology
- (2018) Stanley Weiss Outstanding Dissertation Award, University of Illinois at Urbana–Champaign
- (2016) Hassan Aref Award (research in fluid mechanics), University of Illinois at Urbana–Champaign
- (2015) Alumni Teaching Fellowship, University of Illinois at Urbana–Champaign
- (2010–13) Dean's List, University of Michigan–Dearborn
- (2011) Pi Tau Sigma (honor society, member), University of Michigan–Dearborn

# 6 Research support

## 6.1 Funded grants

### 6.1.1 Current

1. (2025–30) Institutional PI: DOE PSAAP IV PSC DE-NA0004263, "*Center for Multiscale Modeling of Multiphase Combustion*" ($10M, GT Share: $1M, SHB Share: $1M); Director: S. Balachandar (UFlorida)
2. (2025–30) Institutional PI: DOE PSAAP IV FIC DE-NA0004261 "*Center for Information Geometric Mechanics and Optimization*" ($5M, GT Share: $1.2M, SHB Share: $669K) Co-directors: B. Keith (Brown) and F. Schäfer (NYU)
3. (2025–26) PI: DOE DE-NA0003525 (Sandia National Laboratories subcontract), "*Vibrated bubbly flow simulation*" ($113K)
4. (2023–28) co-PI: DOD ONR MURI N00014-23-1-2501, "*Combustion of solid fuels in high enthalpy flow*" ($3.8M) PI: G. Young (Virginia Polytechnic Institute and State University), 7 other co-PIs. SHB Share: $270K Y1–3
5. (2024–27) co-PI: DOD ONR N00014-24-1-2094 "*Multi-scale simulations of combustion in a solid propellant ramjet with embedded reactive metal particles*" ($375K), PI: S. Menon (GT), SHB Share: $188K
6. (2023–27) PI: DOD ARO W911NF-23-10324, "*Investigation and inference of soft material deformation mechanisms unlocked at large speeds, finite deformations, and many cycles*," collaborative with University of Michigan, Jon Estrada. (Total: $835K; SHB Share: $314K)
7. (2022–26) PI: DOD ONR N00014-22-12519, "*Stochastic framework for cavitating flows: mesoscale modeling and acceleration*" ($560K)

### 6.1.2 Completed

1. (2025) PI: Visiting Professor, DOE Pacific Northwest National Laboratory, Subcontract 802894 of DE-AC05-76RL01830, "*Quantum Simulation of Fluid Dynamics*" ($42K)
2. (2024–25) PI: DOE DE-NA0003525 (Sandia National Laboratories subcontract), "*Vibrated bubbly flow simulation*" ($113K)
3. (2024–25) Senior personnel: DARPA HR-0011-472506 "*Squid-inspired nozzles for enhanced efficiency and thrust in rotary propulsors*" ($400K), PI: S. Bhamla (GT), SHB Share: $150K
4. (2024) PI: DOE DE-AC52-07NA27344 (Lawrence Livermore National Laboratories subcontract), "*Accelerated, Compressed, and Regularized Computation of Kinetic-based PDEs*" ($80K)
5. (2023–24) PI: DOE DE-NA0003525 (Sandia National Laboratories subcontract), "*Vibrated bubbly flow simulation*" ($100K)
6. (2023–24) co-PI: DARPA HR-0011-2330006, "*Quantum eigensolvers in fluid-dynamic computations and applications*" ($300K), PI B. Gard (Georgia Tech Research Institute), SHB Share: $100K
7. (2022–23) PI: DOE DE-NA0003525 (Sandia National Laboratories subcontract), "*Vibrated bubbly flow simulation*" ($65K)
8. (2022–23) PI: DOE ORAU Powe, "*A methodologically coherent multi-scale model for multiphase flow*" ($10K)
9. (2022–23) co-PI: GTRI IRAD, "*Quantum optimization for lattice Boltzmann simulation (QOLBS)*" ($40K), PI: B. Gard (Georgia Tech Research Institute)
10. (2022) PI: GT Seed Grant, Forming Teams "*Quantum computing for next-generation engineering simulation*" ($50K)
11. (2022) PI: GTQA DE-0001321I, "*Quantum algorithms for lattice Boltzmann fluid flow simulation*" ($14.5K)

### 6.2 Funded resource and hardware awards

1. (2026–27) PI: AMD University Program (AUP) AI & HPC Cluster (1550 node-hours, 8x MI300X/MI350X GPUs/node)
2. (2026–27) PI: DOE Innovative and Novel Computational Impact on Theory and Experiment (INCITE) Award, "*Exascale-enabled Simulation of Cavitation for Medicine and Beyond*" (700K node hours, OLCF Frontier), co-PIs: T. Colonius, M. Rodriguez, F. Liu, R. Budiardja
3. (2026–27) co-PI: DOE ASCR Leadership Computing Challenge Award (ALCC), "*Multiphase Mixing Induced by Interface Breakup*" (550K node hours, OLCF Frontier), PI: R. M. McMullen (Sandia National Labs)
4. (2025–26) co-PI: DOE ASCR Leadership Computing Challenge Award (ALCC), "*Multiphase Mixing Induced by Interface Breakup*" (495K node hours, OLCF Frontier), PI: R. M. McMullen (Sandia National Labs)
5. (2025) PI: LLNL El Capitan, Full-system dedicated access (11,136 nodes, 44,500+ AMD MI300A APUs; 6 hours post-classification, 2 days pre-classification)
6. (2025) PI: CSCS Alps, Full-system dedicated access (48 hours)
7. (2024–Present) PI: ACCESS-CI Maximize TG-PHY240200, "*Direct simulation of compressible multiphase flow*" (225K GPU Hours, 55K CPU Hours, $119K value)
8. (2024–Present) PI: ACCESS-CI Accelerate TG-PHY210084, "*High-fidelity simulation of high-flowing dispersions*" (3M ACCESS Credits, $24K value)
9. (2021–Present) PI: Oak Ridge National Lab CFD154, Director's Discretionary, "*Accelerated sub-grid multi-component flow physics*" (100K node hours+)
10. (2024–25) PI: Jülich Supercomputing Center, JUPITER Exascale Early Access Program, "*ExaMFlow: Exascale simulation enables multiphase flow simulation at the finest scales*"
11. (2021–23) PI: ACCESS-CI Discovery TG-PHY210084, "*High-fidelity simulation of high-speed flowing dispersions via a stochastic sub-grid model*" (10K GPU Hours, 20K CPU Hours, $7.5K value)
12. (2024) co-PI: Georgia Tech Tech. Fee, "*Next Generation NVIDIA HPC Cluster*" (4x NVIDIA GraceHopper Superchip nodes, $250K)
13. (2022) PI: NVIDIA Academic Hardware Grant Program (4x BlueField-2 E-Series DPU, $12K value)
14. (2022) PI: Georgia Tech Tech. Fee "*ARM HPC Dev Kits for next-generation supercomputing*" (10x NVIDIA ARM HPC Dev. Kits, $240K)
15. (2022) PI: AMD MI200-series GPU Server ($77K value)
16. (2022) PI: NVIDIA Academic Hardware Grant Program (2x A100 80GB PCIe GPUs, $30K value)
17. (2019–20) co-PI: XSEDE TG-CTS120005, "*Advanced immersed boundary and interface-capturing methods for simulations of complex flows*" (9M CPU hours, $71K value)

### 6.3 Other awarded funds

- (2023) PI: SIAM CSE Travel Award ($1K)
- (2023) PI: APS FECS Travel Grant ($350)
- (2022) PI: Stanford CTR Summer Program "*Fast macroscopic forcing for operator recovery via locality and causality with application to compressible and multiphase flow*" ($8K, with F. Schäfer, SHB share: $4K)

## 7 Professional activity

### 7.1 Appointments and memberships

- (2025–Present) OLCF User Group Executive Board, Member
- (2024–Present) University Consortium for Applied Hypersonics (UCAH)
- (2022–Present) Association for Computing Machinery (ACM), Member

- (2021–Present) NATO Science & Technology Organization, Technical Team Member
- (2021–Present) American Institute of Aeronautics & Astronautics (AIAA), Member
- (2015–Present) Society of Industrial and Applied Mathematics (SIAM), Member
- (2014–Present) American Physical Society (APS), Member

## 7.2 Referee

### 7.2.1 Journals and Conferences

- Advances in Engineering Software
- AIAA Journal
- Applied Mathematical Modelling
- Applied Ocean Research
- Computers and Fluids
- Computers and Mathematics with Applications
- Computers in Biology and Medicine
- Computer Methods in Applied Mechanics and Engineering
- Computer Standards & Interfaces
- Communications in Nonlinear Science and Numerical Simulation
- Current Opinion in Chemical Engineering
- European Journal of Mechanics – B/Fluids
- Fluids
- Frontiers in Heat and Mass Transfer
- IEEE International Parallel & Distributed Processing Symposium
- International Journal of High Performance Computing Applications
- International Journal of Multiphase Flow
- International Journal of Offshore and Polar Engineering
- Journal of Computational Physics
- Journal of Computational Science
- Journal of Fluid Mechanics
- Measurement
- Multidiscipline Modeling in Materials and Structures
- Nature Communications Physics
- Ocean Engineering
- PEARC (Practice and Experience in Advanced Research Computing)
- Physical Review A
- Physical Review Applied
- Physical Review E
- Physical Review Fluids
- Physical Review Research
- Physical Review X
- PLOS Computational Biology
- Quantum Information Processing
- Quantum Science and Technology
- SC (International Conference for High Performance Computing, Networking, Storage, and Analysis)
- SIAM Multiscale Modeling & Simulation
- SIAM Scientific Computing
- Soft Matter
- SoftwareX
- Symposium of Naval Hydrodynamics
- Theoretical and Computational Fluid Dynamics

### 7.2.2 Research proposals

Israel Science Foundation, US Department of Defense (Army Research Office), US Department of Energy (INCITE), US National Science Foundation (ENG), ACS Research Funds

# 8 Service and outreach

## 8.1 Georgia Tech

### 8.1.1 Institute-level

- (2021–26) Georgia Tech *HPC Hackathon*, initiator and organizer, recruited sponsors Oak Ridge National Lab and NVIDIA
- (2022–26) Georgia Tech *Scientific Software Engineering Center*, Advisory Board

- (2022–26) PURA Award Reviewer
- (2024) Schmidt Science Polymaths Award Reviewer
- (2022,2023) ORAU Powe Award Reviewer
- (2022) Faculty Search Panel, Professional Development Workshops, Georgia Tech Center for Teaching and Learning

### 8.1.2 College-level

- (2025) CRNCH Summit Organizer (with H. Kim, T. Krishna, D. Mahajan, S. Yu)
- (2024–26) Modeling & Simulation, School of CSE, Area lead
- (2022–26) VIP Team Phoenix–Cluster Competition Team, Faculty advisor
- (2021–26) TSO advisory committee representative
- (2022–24) CSE communication committee
- (2021–24) Seminar series organizer (with F. Schäfer)
- (2023) Computational Mathematics Activity Group (organized by N. Chandramoorthy)
- (2023) CRNCH Summit Panel organizer and moderator (with R. Vuduc)
- (2022) Organizer, Georgia Scientific Computing Symposium (with E. Chow and X. Zhang)
- (2022) Judge, CS Junior Design Capstone Expo
- (2021–22) Graduate student admissions committee

### 8.1.3 Student examination committees

*Ph.D. Thesis Defense*

- (2026) Anand Radhakrishnan (COC CSE)
- (2026) Haocheng Yu (COC CSE)
- (2026) Zhixin (Jack) Song (COS Physics)
- (2025) Liana Hatoum (COE BME)
- (2024) Hohyun Lee (COE ME)
- (2023) Fatima Ezahra Chrit (COE ME)
- (2022) Achyut Panchal (COE AE)
- (2022) Wangwei Lan (COS Physics)

*Ph.D. Thesis Proposal*

- (2026) Anand Radhakrishnan (COC CSE)
- (2024) Micaiah Smith-Pierce (COE AE)
- (2024) Sara Karamati (COC CSE)
- (2023) Liana Hatoum (COE BME)
- (2023) Hohyun Lee (COE ME)
- (2022) Johnie Sublett (COC CSE)

*Ph.D. Qualifying Examination*

- (2025) Jacob Tronge (COC CSE)
- (2025) Mehdi Karimi (COE ME)
- (2025) Sijian Tan (COE AE/CSE)
- (2025) Jasrayman Thind (COE AE/CSE)
- (2024) Srikanth Avasarala (COC CSE)
- (2024) Benjamin Wilfong (COC CSE)
- (2024) Jesus Arias (COC CSE)
- (2024) Lynn Jin (COS Physics)
- (2024) Sijian Tan (COE AE)
- (2023) Ayush Jain (COC CSE)
- (2023) Grayson Harrington (COC CSE)
- (2022) Anand Radhakrishnan (COC CSE)
- (2022) Johnie Sublett (COC CSE)
- (2021) Bradley Baker (COC CSE)
- (2021) Conlain Kelly (COC CSE)
- (2021) Sam Swanson (COC CSE)

*Other*

- (2025) M.S. Thesis proposal; Joseph Walker Brown (COE AE)
- (2025) M.S. Thesis proposal; Filippo Di Benedetto (COE AE)
- (2023) M.S. Thesis defense; Felix Luo (COE AE)
- (2023) M.S. Thesis proposal; Felix Luo (COE AE)

### 8.2 External

- (2026) Breakout session moderator, OLCF User Town Hall, Oak Ridge Leadership Computing Facility
- (2026) Mini-symposium organizer, "Outstanding Problems in and New Solutions for Shock-Laden Flows," US National Congress on Theoretical and Applied Mechanics (USNCTAM) (with F. Schäfer)
- (2026) Mini-symposium organizer, "Cavitation and Bubble Dynamics," US National Congress on Theoretical and Applied Mechanics (USNCTAM) (with J. Yang, M. Rodriguez, J. Estrada)
- (2026) Mini-symposium organizer, "Cavitation and Soft Material Characterization," World Congress on Computational Mechanics – European Congress on Computational Methods in Applied Sciences and Engineering (WCCM–ECCOMAS) (with J. Yang, M. Rodriguez, J. Estrada)
- (2025) Mini-symposium organizer, "Cavitation and Bubble Dynamics," 2025 Society of Engineering Science (SES) Technical Meeting (with M. Rodriguez, J. Estrada, J. Yang)
- (2025) Presenter, APS FECS (Forum for Early Career Scientists) Career Panel: Life in Academia
- (2024) Session chair, International Conference on Theoretical and Applied Mechanics
- (2024) Session chair, International Conference on Numerical Methods in Multiphase Flows
- (2024) Sorting committee, American Physical Society, Division of Fluid Dynamics
- (2022–Present, bi-annual) Panel Referee, ACCESS-CI Maximize
- (2021–Present, annual) Mentor, GPU Hackathons (with Oak Ridge National Lab, NVIDIA, NASA)
- (2021,22,24) Session chair, American Physical Society, Division of Fluid Dynamics
- (2021,23) Poster judge, American Physical Society, Division of Fluid Dynamics
- (2023) Mini-symposium organizer and session chair, "Statistical Approaches to Closure Modeling in Computational Mechanics," IACM Conference on Mechanistic Machine Learning and Digital Engineering for Computational Science, Engineering & Technology (MMLDT-CSET)
- (2023) Session chair, 11th International Conference on Multiphase Flow
- (2022) Supercomputing (SC) Mentor (via Mentor–Protege program)
- (2022) Supercomputing (SC) Early Career Program
- (2021–22) Research mentor, XSEDE EMPOWER (Expert Mentoring Producing Opportunities for Work, Education, and Research; program received HPCwire 2021 Editors' Choice Award in Workforce Diversity and Inclusion Leadership)
- (2021) Mini-symposium organizer and session chair, "Machine learning for multiphase flows," IACM Conference on Mechanistic Machine Learning and Digital Twins for Computational Science, Engineering & Technology (MMLDT-CSET)
- (2020) Research mentor, Schmidt Academy for Software Engineering
- (2019) Research mentor, WAVE undergraduate research program for under-represented students, Caltech
- (2015, 2016) Judge, Illinois State-wide Math Competition
- (2014) Organizer, Science Night, Illinois Middle Schools

## 9 Media

- (2026) What Does OpenAI's Blockbuster Mathematics Proof Mean for the Real World?, by Adam Kovac; Scientific American [LINK]
- (2026) Apple's Neural Engine Holds No More Secrets Thanks to a University Study (in French), by Alexandre Godard; iPhoneSoft [LINK]
  - Also discussed by: Breadboardhub, smolnero
- (2026) Apple's Neural Engine: Architecture, Programming, and Performance Fully Documented for the First Time (in German), by Markus Seyfferth; Dr. Web [LINK]

- (2025) El Capitan's Breakthrough CFD Run and the Shift to Quantum Ready Engineering; BQP [LINK]
- (2025) Record-Breaking Simulation Boosts Rocket Science and Supercomputing to New Limits; Georgia Tech [LINK]
- (2025) Gordon Bell Prize Winners Push Open Science Boundaries With NVIDIA-Powered Supercomputers; NVIDIA [LINK]
- (2025) Gordon Bell finalist team pushes scale of rocket simulation on El Capitan; LLNL [LINK]
  - Also propagated by: HPCwire, Interesting Engineering, Newswise
- (2025) Tracking the fiery path of particles: UF leads combustion research; UFlorida [LINK]
- (2025) It Is Rocket Science; NCSA [LINK]
  - Also propagated by: AIwire, EurekAlert!, Quantum Zeitgeist
- (2025) Fluid flow simulation on Frontier earns Gordon Bell finalist selection; ORNL [LINK]
  - Also propagated by: HPCwire, insideHPC, OLCF
- (2025) Users of Oak Ridge Leadership Computing Facility Gather for 21st Annual Meeting; OLCF [LINK]
- (2025) Shock Treatment for CFD Simulations; OLCF [LINK]
  - Also propagated by: HPCwire, insideHPC
- (2025) GT Supports Launch of Europe's Fastest Supercomputer; Georgia Tech [LINK]
  - Also propagated by: TechXplore, Interesting Engineering
- (2025) The OLCF's Problem Busters; OLCF [LINK]
- (2024) Featured: Art of HPC: Red Blood Cells Flowing Through a Microaneurysm Using High-Performance Computing; Georgia Tech [LINK]
- (2024) Researchers Blazing New Trails with Superchip Named After Computing Pioneer; Georgia Tech [LINK]
- (2023) GTRI, Georgia Tech Use Quantum Computing to Optimize CFD Applications; Georgia Tech Research Institute [LINK]
- (2023) Group Optimizes Fluid Dynamics Simulator on World's Fastest Supercomputer; Georgia Tech [LINK]
- (2023) Researchers Optimize HPC Software at Interdisciplinary Hackathon; Georgia Tech [LINK]
- (2022) New Hardware Brings Students Closer to Exascale Computing; Georgia Tech [LINK]
- (2022) Faculty Receives New GPUs for Fluid Dynamics and Machine Learning Research; Georgia Tech [LINK]

# 10 Publications

*Bolding indicates advised or co-advised students and postdocs.*

## 10.1 Preprints

[PP2] **Chu, T.**, **Wilfong, B.**, Koehler, T., McMullen, R. M., Bryngelson, S. H., (2026). "Coupled Rayleigh–Taylor and Faraday instabilities in vertically vibrated cylindrical containers". arXiv:2607.28932. DOI: 10.48550/arXiv.2607.28932.

[PP1] Eyob, B., **Arias, J.**, Bryngelson, S. H., Schäfer, F., (2026). "Discontinuous Galerkin semidiscretization of the information geometric regularized compressible Euler equations". arXiv:2608.02223. DOI: 10.48550/arXiv.2608.02223.

## 10.2 Archival, heavily refereed papers

[P50] Bryngelson, S. H. (2026). "ANEForge: Python for direct computation on the Apple Neural Engine". *Software: Practice and Experience*. In press. DOI: 10.1002/spe.70124.

[P49] Bryngelson, S. H. (2026). "Fast integration method for averaging polydisperse bubble population dynamics". *Computers & Fluids* **304**, 106877. DOI: 10.1016/j.compfluid.2025.106877.

[P48] Bryngelson, S. H., Fox, R. O., Laurent, F., (2026). "Fourth-order HyQMOM closures for multidimensional kinetic equations". *Journal of Computational Physics* **566**, 115242. DOI: 10.1016/j.jcp.2026.115242.

[P47] **Chu, T.**, Beckett, J., Zhu, Z., Estrada, J. B., Bryngelson, S. H., (2026). "Limits of constant-parameter constitutive models for hydrogels under inertial cavitation". *Soft Matter*. DOI: 10.1039/D6SM00547K.

[P46] **Chu, T.**, Estrada, J. B, Bryngelson, S. H., (2026). "Accelerating Bayesian optimal experimental design via local radial basis functions: Application to soft material characterization". *Journal of Computational Physics* **562**, 115002. DOI: 10.1016/j.jcp.2026.115002.

[P45] Cisneros-Garibay, E., **Le Berre, H.**, **Adam, D.**, Bryngelson, S. H., Freund, J. B., (2026). "Pyrometheus: Symbolic abstractions for XPU and automatically differentiated computation of combustion kinetics and thermodynamics". *Computer Physics Communications* **320**, 109987. DOI: 10.1016/j.cpc.2025.109987.

[P44] Goldstein, B. I., Ramsey, D. N., **Yu, H.**, Bryngelson, S. H., Ahuja, K. K., (2026). "Performance of Sharply Bent Acoustic Resonators at High Sound Levels". *AIAA Journal* **64** 8, 4719–4728. DOI: 10.2514/1.J066497.

[P43] Jawetz, C. L., **Song, Z.**, Alexeev, A., Bryngelson, S. H., (2026). "Quantum lattice Boltzmann algorithm for heat transfer with phase change". *Quantum Science and Technology* **11** 3, 035032. DOI: 10.1088/2058-9565/ae7b7c.

[P42] **Lee, M.**, **Song, Z.**, **Kocherla, S.**, Adams, A., Alexeev, A., Bryngelson, S. H., (2026). "A multiple-circuit approach to quantum resource reduction with application to the quantum lattice Boltzmann method". *Future Generation Computing Systems* **174**, 107975. DOI: 10.1016/j.future.2025.107975.

[P41] **Radhakrishnan, A.**, Bryngelson, S. H., (2026). "A sub-grid-scale model for polydisperse bubbly flows with heat and mass transfer". *International Journal of Multiphase Flow*. In press. DOI: 10.48550/arXiv.2607.27426.

[P40] **Radhakrishnan, A.**, **Wilfong, B.**, Bryngelson, S. H., Schäfer, F., (2026). "Shocks without shock capturing: Information geometric regularization of finite volume methods for Navier-Stokes-like problems". *Computer Methods in Applied Mechanics and Engineering* **461**, 119216. DOI: 10.1016/j.cma.2026.119216.

[P39] Sanchez, V., Remillard, S., Abeid, B. A., Bu, L., Bryngelson, S. H., Yang, J., Estrada, J. B., Rodriguez Jr. M., (2026). "Hierarchical Bayesian constitutive model selection for high-strain-rate soft material characterization". *Soft Matter* **22**, 3207–3223. DOI: 10.1039/D5SM01193K.

[P38] **Song, Z.**, Ren, H., **Lee, M.**, Gard, B., Renaud, N., Bryngelson, S. H., (2026). "Hadamard random forest for reconstructing real valued pure quantum states". *Communications Physics*. DOI: 10.1038/s42005-026-02717-1.

[P37] Vaca-Revelo, D., **Wilfong, B.**, Bryngelson, S. H., Gnanaskandan, A., (2026). "Hardware-Accelerated Phase-Averaging for Cavitating Bubbly Flows". *International Journal of Multiphase Flow* **199**, 105674. DOI: 10.1016/j.ijmultiphaseflow.2026.105674.

[P36] **Wilfong, B.**, **Le Berre, H.**, **Radhakrishnan, A.**, **Gupta, A.**, **Vickers, D. J.**, Vaca-Revelo, D., **Adam, D.**, **Yu, H.**, **Lee, H.**, Chreim, J. R., Carcana Barbosa, M., Zhang, Y., Cisneros-Garibay, E., Gnanaskandan, A., Rodriguez Jr. M., Budiardja, R. D., Abbott, S., Colonius, T., Bryngelson, S. H., (2026). "MFC 5.0: An exascale many-physics flow solver". *Computer Physics Communications* **322**, 110055. DOI: 10.1016/j.cpc.2026.110055.

[P35] **Yu, H.**, Ahuja, K., Sankar, L. L., Bryngelson, S. H., (2026). "Transmission of High-Amplitude Sound through Leakages of Ill-fitting Earplugs". *International Journal of Aeroacoustics*. In press. DOI: 10.1177/1475472X261473636.

[P34] ***Yu, H.**, ***Chu, T.**, Bryngelson, S. H., (2026). "Energy dissipation mechanisms in an acoustically driven slit". *Journal of Fluid Mechanics* **1040**. *Equal contribution, A47. DOI: 10.1017/jfm.2026.11896.

[P33] **Chu, T.**, Estrada, J. B., Bryngelson, S. H., (2025). "Bayesian optimal design accelerates discovery of soft material properties from bubble dynamics". *Computational Mechanics* **76**, 431–447. DOI: 10.1007/s00466-025-02606-4.

[P32] **Chu, T.**, **Wilfong, B.**, Koehler, T., McMullen, R. M., Bryngelson, S. H., (2025). "Competing mechanisms at vibrated interfaces of density-stratified fluids". *Physical Review Fluids* **10**, 093904. DOI: 10.1103/r9b3-psg4.

[P31] **Song, Z.**, Deaton, R., Gard, B., Bryngelson, S. H., (2025). "Incompressible Navier–Stokes solve on noisy quantum hardware via a hybrid quantum–classical scheme". *Computers & Fluids* **288**, 106507. DOI: 10.1016/j.compfluid.2024.106507.

[P30] **Wilfong, B.**, **Radhakrishnan, A.**, **Le Berre, H. A.**, **Prathi, T.**, Abbott, S., Bryngelson, S. H., (2025). "Testing and benchmarking emerging supercomputers via the MFC flow solver". *SC Workshops '25: Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 669–677. DOI: 10.1145/3731599.3767424.

[P29] **Wilfong, B.**, **Radhakrishnan, A.**, **Le Berre, H.**, **Vickers, D. J.**, **Prathi, T.**, Tselepidis, N., Dorschner, B., Budiardja, R., Cornille, B., Abbott, S., *Schäfer, F., *Bryngelson, S. H., (2025). "Simulating many-engine spacecraft: Exceeding 1 quadrillion degrees of freedom via information geometric regularization". *Proceedings of SC '25: The International Conference for High Performance Computing, Networking, Storage and Analysis*. *Equal contribution, 14–24. DOI: 10.1145/3712285.3771783.

[P28] Zhu, Z., Remillard, S., Abeid, B. A., Frolkin, D., Bryngelson, S. H., Yang, J., Rodriguez Jr. M., Estrada, J. B., (2025). "Parsimonious inertial cavitation rheometry via bubble collapse time". *Soft Matter* **21** 34, 6717–6734. DOI: 10.1039/D5SM00397K.

[P27] **Bati, A.**, Bryngelson, S. H., (2024). "RoseNNa: A performant, portable library for neural network inference with application to computational fluid dynamics". *Computer Physics Communications* **296**, 109052. DOI: 10.1016/j.cpc.2023.109052.

[P26] *Bryngelson, S. H., *Schäfer, F., Liu, J., Mani, A., (2024). "Fast Macroscopic Forcing Method". *Journal of Computational Physics* **499**. *Equal contribution, 112721. DOI: 10.1016/j.jcp.2023.112721.

[P25] **Kocherla, S.**, **Song, Z.**, **Chrit, F. E.**, Gard, B., Dumitrescu, E. F., Alexeev, A., Bryngelson, S. H., (2024). "Fully quantum algorithm for mesoscale fluid simulations with application to partial differential equations". *AVS Quantum Science* **6**, 033806. DOI: 10.1116/5.0217675.

[P24] Liu, J., Schäfer, F., Bryngelson, S. H., Zaki, T. A., Mani, A., (2024). "Adjoint-based computation of nonlocal eddy viscosity in turbulent channel flow". *Physical Review Fluids* **9**, 094606. DOI: 10.1103/PhysRevFluids.9.094606.

[P23] **Radhakrishnan, A.**, **Le Berre, H.**, **Wilfong, B.**, Spratt, J.-S., Rodriguez Jr. M., Colonius, T., Bryngelson, S. H., (2024). "Method for portable, scalable, and performant GPU-accelerated simulation of multiphase compressible flow". *Computer Physics Communications* **302**, 109238. DOI: 10.1016/j.cpc.2024.109238.

[P22] **Sinha, A.**, Bryngelson, S. H., (2024). "Neural networks can be FLOP-efficient integrators of 1D oscillatory integrands". *Transactions on Machine Learning Research*. ISSN: 2835-8856.

[P21] **Wilfong, B.**, **Radhakrishnan, A.**, **Le Berre, H. A.**, Abbott, S., Budiardja, R. D., Bryngelson, S. H., (2024). "OpenACC offloading of the MFC compressible multiphase flow solver on AMD and NVIDIA GPUs". *SC24-W: Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 1923–1933. DOI: 10.1109/SCW63240.2024.00242.

[P20] Bryngelson, S. H., Fox, R. O., Colonius, T., (2023). "Conditional moment methods for polydisperse cavitating flows". *Journal of Computational Physics* **477**, 111917. DOI: 10.1016/j.jcp.2023.111917.

[P19] Elwasif, W., Bastrakov, S., Bryngelson, S. H., Bussmann, M., Chandrasekaran, S., Ciorba, F., Clark, M. A., Debus, A., Godoy, W., Hagerty, N., Hammond, J., Hardy, D., Harris, J. A., Hernandez, O., Joo, B., Keller, S., Kent, P., **Le Berre, H.**, Lebrun-Grandie, D., MacCarthy, E., Vergara, V. G. M., Messer, B., Miller, R., Oral, S., Piccinali, J.-G., **Radhakrishnan, A.**, Simsek, O., Spiga, F., Steiniger, K., Stephan, J., Stone, J. E., Trott, C., Widera, R., Young, J., (2023). "Early application experiences on a modern GPU-accelerated Arm-based HPC platform". *HPC Asia '23*. International Workshop on Arm-based HPC: Practice and Experience (IWAHPCE). Singapore, 35–49. DOI: 10.1145/3581576.3581621.

[P18] Firouznia, M., Bryngelson, S. H., Saintillan, D., (2023). "A spectral boundary integral method for simulating electrohydrodynamic flows in viscous drops". *Journal of Computational Physics* **489**, 112248. DOI: 10.1016/j.jcp.2023.112248.

[P17] **Panchal, A.**, Bryngelson, S. H., Menon, S., (2023). "A seven-equation diffused interface method for resolved multiphase flows". *Journal of Computational Physics* **475**, 111870. DOI: 10.1016/j.jcp.2022.111870.

[P16] **Zeng, Q.**, **Kothari, Y.**, Bryngelson, S. H., Schäfer, F., (2023). "Competitive physics informed networks". *International Conference on Learning Representations (ICLR)*. arXiv:2204.11144. Kigali, Rwanda.

[P15] Charalampopoulos, A., Bryngelson, S. H., Colonius, T., Sapsis, T. P., (2022). "Hybrid quadrature moment method for accurate and stable representation of non-Gaussian processes and their dynamics". *Philosophical Transactions of the Royal Society A* **380** 2229. DOI: 10.1098/rsta.2021.0209.

[P14] Bryngelson, S. H., Schmidmayer, K., Coralic, V., Maeda, K., Meng, J., Colonius, T., (2021). "MFC: An open-source high-order multi-component, multi-phase, and multi-scale compressible flow solver". *Computer Physics Communications* **266**, 107396. DOI: 10.1016/j.cpc.2020.107396.

[P13] Spratt, J.-S., Rodriguez, M., Schmidmayer, K., Bryngelson, S. H., Yang, J., Franck, C., Colonius, T., (2021). "Characterizing viscoelastic materials via ensemble-based data assimilation of bubble collapse observations". *Journal of the Mechanics and Physics of Solids* **152**, 104455. DOI: 10.1016/j.jmps.2021.104455.

[P12] Bryngelson, S. H., Charalampopoulos, A., Sapsis, T. P., Colonius, T., (2020). "A Gaussian moment method and its augmentation via LSTM recurrent neural networks for the statistics of cavitating bubble populations". *International Journal of Multiphase Flow* **127**, 103262. DOI: 10.1016/j.ijmultiphaseflow.2020.103262.

[P11] Bryngelson, S. H., Colonius, T., (2020). "Simulation of humpback whale bubble-net feeding models". *Journal of the Acoustical Society of America* **147** 2, 1126–1135. DOI: 10.1121/10.0000746.

[P10] Bryngelson, S. H., Colonius, T., Fox, R. O., (2020). "QBMMlib: A library of quadrature-based moment methods". *SoftwareX* **12**, 100615. DOI: 10.1016/j.softx.2020.100615.

[P9] Schmidmayer, K., Bryngelson, S. H., Colonius, T., (2020). "An assessment of multicomponent flow models and interface capturing schemes for spherical bubble dynamics". *Journal of Computational Physics* **402**, 109080. DOI: 10.1016/j.jcp.2019.109080.

[P8] Trummler, T., Bryngelson, S. H., Schmidmayer, K., Schmidt, S. J., Colonius, T., Adams, N. A., (2020). "Near-surface dynamics of a gas bubble collapsing above a crevice". *Journal of Fluid Mechanics* **899**, A16. DOI: 10.1017/jfm.2020.432.

[P7] Bryngelson, S. H., Freund, J. B., (2019). "Non-modal Floquet stability of a capsule in large amplitude oscillatory extension". *European Journal of Mechanics B/Fluids* **77**, 171–176. DOI: 10.1016/j.euromechflu.2019.04.012.

[P6] Bryngelson, S. H., Guéniat, F., Freund, J. B., (2019). "Irregular dynamics of cellular blood flow in a model microvessel". *Physical Review E* **100**, 012203. DOI: 10.1103/PhysRevE.100.012203.

[P5] Bryngelson, S. H., Schmidmayer, K., Colonius, T., (2019). "A quantitative comparison of phase-averaged models for bubbly, cavitating flows". *International Journal of Multiphase Flow* **115**, 137–143. DOI: 10.1016/j.ijmultiphaseflow.2019.03.028.

[P4] Bryngelson, S. H., Freund, J. B., (2018). "Floquet stability analysis of capsules in viscous shear flow". *Joural of Fluid Mechanics* **852**, 663–677. DOI: 10.1017/jfm.2018.574.

[P3] Bryngelson, S. H., Freund, J. B., (2018). "Global stability of flowing red blood cell trains". *Physical Review Fluids* **3** 7, 073101. DOI: 10.1103/PhysRevFluids.3.073101.

[P2] Bryngelson, S. H., Freund, J. B., (2016). "Buckling and its effect on the confined flow of a model capsule suspension". *Rheologica Acta* **55** 6, 451–464. DOI: 10.1007/s00397-015-0900-9.

[P1] Bryngelson, S. H., Freund, J. B., (2016). "Capsule-train stability". *Physical Review Fluids* **1** 3, 033201. DOI: 10.1103/PhysRevFluids.1.033201.

## 10.3 Conference papers

[C26] **Chu, T.**, **Wilfong, B.**, Koehler, T., McMullen, R. M., Bryngelson, S. H., (2026). "Linear and numerical analysis of interfacial instabilities in vibrated cylindrical containers". *AIAA AVIATION 2026 Forum*, 4363. DOI: 10.2514/6.2026-4363.

[C25] Cisneros-Garibay, E., **Adam, D.**, Bryngelson, S. H., (2026). "A symbolic computational abstraction of chemistry libraries". *AIAA SCITECH 2026 Forum*, 1812. DOI: 10.2514/6.2026-1812.

[C24] Goldstein, B., Ramsey, D. N., **Yu, H.**, Bryngelson, S. H., Ahuja, K. K., (2025). "Performance of sharply bent acoustic resonators at high sound levels". *AIAA Aviation Forum and ASCEND 2025*, 3748. DOI: 10.2514/6.2025-3748.

[C23] **Radhakrishnan, A.**, Schäfer, F., Bryngelson, S. H., (2025). "Solving diffuse interface models without tracking or dissipation". *12th International Conference on Multiphase Flow*. Toulouse, France.

[C22] Carcana Barbosa, M., Yang, J., Estrada, J. B., Bryngelson, S. H., Rodriguez Jr. M., (2024). "Numerical simulations of inertial bubble collapse near a hyperelastic object". *11th International Symposium on Cavitation*. Crete, Greece.

[C21] **Radhakrishnan, A.**, Bryngelson, S. H., (2024). "A statistics-based sub-grid model for cavitation inception and its application to complex flows". *11th International Symposium on Cavitation*. Crete, Greece.

[C20] **Radhakrishnan, A.**, Bryngelson, S. H., (2024). "A stochastic representation of sub-grid bubble dynamics toward the modeling of cavitation inception". *35th Symposium on Naval Hydrodynamics*. Nantes, France.

[C19] **Radhakrishnan, A.**, Bryngelson, S. H., (2024). "Stochastic computational methods for cavitation inception". *XXVI International Congress of Theoretical and Applied Mechanics*. Daegu, South Korea.

[C18] **Wilfong, B.**, McMullen, R. M., Koehler, T., Bryngelson, S. H., (2024). "Instability of two-species interfaces via vibration". *AIAA Aviation Forum and ASCEND 2024*, 4480. DOI: 10.2514/6.2024-4480.

[C17] **Wilfong, B.**, **Radhakrishnan, A.**, Bryngelson, S. H., (2024). "Multiphase flow numerics: Perspectives from exascale simulation". *5th International Conference on Numerical Methods in Multiphase Flows (ICNMMF5)*. Reykjavik, Iceland.

[C16] **Yu, H.**, Ahuja, K. K., Sankar, L. N., Bryngelson, S. H., (2024). "Numerical investigation of leakage of high-amplitude sound inill-fitting earplugs". *AIAA Aviation Forum and ASCEND 2024*, 4391. DOI: 10.2514/6.2024-4391.

[C15] **Le Berre, H. A.**, **Radhakrishnan, A.**, Bryngelson, S. H., (2023). "Fast simulation of multiphase compressible flows thorugh GPU acceleration". *11th International Conference on Multiphase Flow*. Kobe, Japan.

[C14] **Radhakrishnan, A.**, **Le Berre, H. A.**, Bryngelson, S. H., Chreim, J. R., Colonius, T., (2023). "A stochastic computational method for bubbly flows with first steps towards representing inception". *11th International Conference on Multiphase Flow*. Kobe, Japan.

[C13] Bryngelson, S. H., Charalampopoulos, A., Sapsis, T. P., Fox, R. O., Colonius, T., (2022). "Representing statistics of dispersions via moment methods and recurrent neural networks with application to cavitating bubbles". *34th Symposium on Naval Hydrodynamics*. Washington D.C., USA.

[C12] Bryngelson, S. H., Schäfer, F., Liu, J., Mani, A., (2022). "Fast Macroscopic Forcing Method". *Center for Turbulence Research, Proceedings of the Summer Program*. Stanford, CA, USA.

[C11] **Radhakrishnan, A.**, **Le Berre, H.**, Bryngelson, S. H., (2022). "Scalable GPU accelerated simulation of multiphase compressible flow". *The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*. Dallas, TX, USA.

[C10] Rodriguez, M., Bryngelson, S. H., Colonius, T., (2022). "Bubble dynamics with phase change near a compliant object". *34th Symposium on Naval Hydrodynamics*. Washington D.C., USA.

[C9] Bryngelson, S. H., Colonius, T., (2021). "Closure of phase-averaged bubbly, cavitating flow models". *XXV International Congress of Theoretical and Applied Mechanics*. Milano, Italy. URL: https://vimeo.com/640932583/0ae772bf00.

[C8] Bryngelson, S. H., O'Meally, F., Colonius, T., Fox, R. O., (2021). "Conditional moment method for fully-coupled phase-averaged cavitation models". *11th International Symposium on Cavitation*. Daejeon, Korea. URL: https://vimeo.com/640931949/a6cd12fc05.

[C7] Rodriguez, M., Bryngelson, S. H., Cao, S., Colonius, T., (2021). "A unified Eulerian multiphase framework for fluid-structure interaction problems including cavitation". *XXV International Congress of Theoretical and Applied Mechanics*. Milano, Italy.

[C6] Rodriguez, M., Bryngelson, S. H., Cao, S., Colonius, T., (2021). "Acoustically-induced bubble growth and phase change dynamics near compliant surfaces". *11th International Symposium on Cavitation*. Daejeon, Korea.

[C5] Spratt, J.-S., Rodriguez, M., Bryngelson, S. H., Cao, S., Colonius, T., (2021). "Eulerian framework for bubble-cloud-kidney stone interaction". *11th International Symposium on Cavitation*. Daejeon, Korea.

[C4] Bryngelson, S. H., Colonius, T., (2020). "Phase- and mixture-averaged techniques for general bubbly flows". *33rd Symposium on Naval Hydrodynamics*. Osaka, Japan. URL: https://vimeo.com/640930931/6e57ccfd89.

[C3] Bryngelson, S. H., Colonius, T., (2019). "A comparison of ensemble- and volume-averaged bubbly flow models". *10th International Conference on Multiphase Flow*. Rio de Janeiro, Brazil.

[C2] Bryngelson, S. H., Freund, J. B., (2016). "Buckling and the rheology of an elastic capsule suspension". *XXIV International Congress of Theoretical and Applied Mechanics*. Montreal, Canada.

[C1] Freund, J. B., Bryngelson, S. H., (2016). "The stability of flowing trains of confined red blood cells". *XXIV International Congress of Theoretical and Applied Mechanics*. Montreal, Canada.

### 10.4 Other published content

[O4] Bryngelson, S. H. (2026). *Apple Neural Engine: Architecture, Programming, and Performance*. arXiv.2606.22283. DOI: 10.48550/arXiv.2606.22283.

[O3] Shahane, S., Chammas, S., Bezgin, D. A., Buhendwa, A. B., Schmidt, S. J., Adams, N. A., Bryngelson, S. H., Chen, Y.-F., Wang, Q., Sha, F., Zepeda-Núñez, L., (2024). *Rational-WENO: A lightweight, physically-consistent three-point weighted essentially non-oscillatory scheme*. arXiv.2409.09217. DOI: 10.48550/arXiv.2409.09217.

[O2] Bryngelson, S. H., Pantano, C., Bodony, D., Freund, J. B., (2018). *Adjoint-based sensitivity for flows with shocks*. Technical Report, XPACC.

[O1] Bryngelson, S. H. (2017). "Stability and transition of capsule-flow systems". Ph.D. Thesis. University of Illinois at Urbana–Champaign.

## 11 Talks

### 11.1 Invited talks

[I56] Apple, *Apple Silicon Engineering*, (2026).

[I55] Oak Ridge National Laboratory, *Oak Ridge Leadership Computing Facility Science Engagement and HPC Operations*, (2026).

[I54] Oak Ridge National Laboratory, *OLCF User Conference Call*, (2026).

[I53] Oak Ridge National Laboratory, *OLCF User Summit*, (2026).

[I52] Royal Aeronautical Society, *Aerodynamics Workshop 2026*, (2026).

[I51] University of Michigan, *MICDE Predictive Science Symposium*, (2026).

[I50] University of Tennessee Space Institute, *Mechanical and Aerospace Seminar*, (2026).

[I49] University of Texas at Austin, *Oden Institute Babuška Forum*, (2026).

[I48] University of Texas at Austin, *Oden Institute Seminar*, (2026).

[I47] Acoustical Society of America, *Invited conference talk*, (2025).

[I46] Boeing, *CFD Community of Excellence Lecture Series*, (2025).

[I45] Courant Institute of Mathematical Sciences, *Computational Mathematics and Scientific Computing Seminar*, (2025).

[I44] DOE Pacific Northwest National Laboratory, *Environmental Technology Seminar*, (2025).

[I43] Georgia Tech IDEaS, *Cyberinfrastructure and Services for Science & Engineering Workshop*, (2025).

[I42] Oak Ridge Leadership Computing Facility (OLCF), *2025 User Meeting*, (2025).

[I41] Open Accelerated Computing (OAC) Summit, (2025).

[I40] SC25, *Gordon Bell Finalist Presentation*, (2025).

[I39] SC25, *NVIDIA Booth Invited Talk*, (2025).

[I38] SC25, *OpenACC User Group Invited Talk*, (2025).

[I37] University of California, Berkeley, *Department of Mechanical Engineering Seminar Series*, (2025).

[I36] University of Florida, *Institute for Computational Engineering Seminar*, (2025).

[I35] University of Washington, *Department of Aerospace Engineering Seminar Series*, (2025).

[I34] Center for Research into Novel Computing Hierarchies, *CRNCH Summit*, (2024).

[I33] Naval Surface Warfare Center, Carderock Division, *Quantum Science Seminar*, (2024).

[I32] Supercomputing (SC) 2024, *ARTHPC Creates*, (2024).

[I31] U.S. Navel Research Laboratory, *Computational Physics & Fluid Dynamics Seminar*, (2024).

[I30] Bayer AG, *Field Data Science Seminar*, (2023).

[I29] Georgia Institute of Technology, *George W. Woodruff School of Mechanical Engineering Seminar Series*, (2023).

[I28] Google Research, *Applied Science Seminar Series*, (2023).

[I27] Lawrence Livermore National Laboratory, *Data-driven Physics Simulation Webinar*, (2023). URL: https://www.youtube.com/watch?v=zm-iF1FtkLE.

[I26] OpenACC Webinar, (2023). URL: https://www.youtube.com/watch?v=S0gRVIkNYPg.

[I25] University of Illinois at Urbana–Champaign, *Mechanical Science and Engineering Fluids Seminar*, (2023).

[I24] Arizona State University, *Fluids Seminar*, (2022).

[I23] Brown University, *Center for Fluid Mechanics, Applied Math and Engineering*, (2022).

[I22] Center for Research into Novel Computing Hierarchies, *CRNCH Summit*, (2022). URL: https://mediaspace.gatech.edu/media/CRNCH+Summit+2022+-+Spencer+Brynglesonn+-+Quantum+Computing+for+Continuum+Mechanics/1_23u8ou36.

[I21] Emory University, *Scientific Computing Seminar Series*, (2022).

[I20] Georgia Institute of Technology, *Applied and Computational Math Seminar Series*, (2022).

[I19] Georgia Institute of Technology, *Daniel Guggenheim School of Aerospace Engineering Seminar Series*, (2022). URL: https://vimeo.com/759713173/12ef9a0220.

[I18] Georgia Scientific Computing Symposium, (2022).

[I17] Massachusetts Institute of Technology, *SAND Group*, (2022).

[I16] Office of Naval Research, *Basic Research Challenge Guest Talks*, (2022).

[I15] California Institute of Technology, *Mechanical and Civil Engineering Seminar Series*, (2021).

[I14] OpenACC Annual Summit, (2021). URL: https://youtu.be/DgX6ssX2yrg.

[I13] University of California, San Diego, *Fluid Mechanics, Combustion, & Engineering Physics Seminar Series*, (2021). URL: https://vimeo.com/640930056/b1a6c0dc62.

[I12] Georgia Institute of Technology, *Computational Science & Engineering Seminar Series*, (2020).

[I11] Massachusetts Institute of Technology, *SAND Group*, (2019).

[I10] University of Michigan–Ann Arbor, *Mechanical Engineering Seminar Series*, (2019).

[I9] University of Michigan–Dearborn, *Mechanical Engineering Seminar Series*, (2019).

[I8] University of Utah, *Mechanical Engineering Seminar Series*, (2019).

[I7] University of Vermont, *Mechanical Engineering Seminar Series*, (2019).

[I6] University of Washington, *Mechanical Engineering Seminar Series*, (2019).

[I5] California Institute of Technology, *Computational Flow Physics Group*, (2018).

[I4] California Institute of Technology, *Flow Mechanics Research Conference*, (2018).

[I3] ETH Zurich, *Computational Science & Engineering Lab*, (2017).

[I2] University of Illinois at Urbana–Champaign, *Fluid Mechanics Seminar*, (2017).

[I1] University of Illinois at Urbana–Champaign, *Biology Interest Group*, (2015).

## 11.2 Conference presentations

[T81] Bryngelson, S. H., **Prathi, T.**, **Radhakrishnan, A.**, **Vickers, D.**, **Wilfong, B.,** (2026). "Compressible flow with a free lunch: Simulating 1 quadrillion degrees of freedom via regularization without loss of accuracy". *SIAM Conference on Parallel Processing for Scientific Computing (PP26)*.

[T80] Bryngelson, S. H., **Radhakrishnan, A.**, **Wilfong, B.,** (2026). "Extreme-scale simulation of many-engine spacecraft via regularization". *US National Congress on Theoretical and Applied Mechanics*.

[T79] Bryngelson, S. H., **Radhakrishnan, A.**, **Wilfong, B.**, Schäfer, F., (2026). "Shocks without tracking or capturing: Exceeding 1 quadrillion degrees of freedom via inviscid regularization of the compressible Navier–Stokes". *American Physical Society March Meeting*.

[T78] **Chu, T.**, Beckett, J., Zhu, Z., Estrada, J. B., Bryngelson, S. H., (2026). "Constitutive property evolution during laser-induced inertial cavitation". *US National Congress on Theoretical and Applied Mechanics*.

[T77] **Song, Z.**, Ren, H., **Lee, M.**, Gard, B., Renaud, N., Bryngelson, S. H., (2026). "Efficient reconstruction for real-valued quantum states with applications to quantum linear system algorithms". *American Physical Society March Meeting*.

[T76] Bryngelson, S. H., **Radhakrishnan, A.**, **Wilfong, B.,** (2025). "Solving Navier–Stokes-like problems with discontinuous features via information geometric regularization". *UNCG PDE Conference 2025*.

[T75] Bryngelson, S. H., **Wilfong, B.**, **Radhakrishnan, A.**, Schäfer, F., (2025). "Simulating shock-laden and multiphase flows interface without tracking or dissipation". *Americal Physical Society, Division of Fluid Dynamics*.

[T74] Bryngelson, S. H. (2025). "Efficient computation of high-amplitude acoustics in the body: From ultrasound to noise". *188th Meeting of the Acoustical Society of America joint with 25th International Congress on Acoustics*.

[T73] Bryngelson, S. H. (2025). "High-fidelity simulation of shock-dominated high-Mach flows via information geometric regularization". *University Consortium for Applied Hypersonics (UCAH) Forum 2025*.

[T72] Bryngelson, S. H. (2025). "Shocks without shock-capturing: Extreme scale solutions to the compressible Navier–Stokes equations via inviscid geometric regularization". *Algorithms For Multiphysics Models In The Post-Moore's Law Era Workshop*.

[T71] Bryngelson, S. H., Schäfer, F., (2025). "Shocks without shock-capturing: Extreme scale solutions to the compressible Navier–Stokes equations via inviscid geometric regularization". *The Third Joint SIAM/CAIMS Annual Meetings (AN25)*.

[T70] Bryngelson, S. H., **Wilfong, B.**, **Radhakrishnan, A.,** (2025). "Numerics for diverse exascale platforms with application to multi-phase and species flow". *18th U.S. National Congress on Computational Mechanics (USNCCM)*.

[T69] **Chu, T.**, Beckett, J., Abeid, B., Estrada, J. B., Bryngelson, S. H., (2025). "Bayesian optimal design accelerates discovery of material properties from bubble dynamics". *SIAM Computational Science and Engineering*.

[T68] **Chu, T.**, **Wilfong, B.**, Koehler, T., McMullen, R., Bryngelson, S. H., (2025). "Competing mechanisms at vibrated interfaces of density-contrast fluids". *Americal Physical Society, Division of Fluid Dynamics*.

[T67] Cisneros-Garibay, E., **Adam, D.**, Bryngelson, S. H., (2025). "Symbolic computational representations of combustion thermochemistry". *International Conference on Numerical Combustion*.

[T66] **Radhakrishnan, A.**, **Le Berre, H.**, **Wilfong, B.**, Budiardja, R., Abbott, S., Bryngelson, S. H., (2025). "Compressible flow simulation on Frontier and El Capitan architectures". *APS Global Physics Summit*.

[T65] Sanchez, V., Remillard, S., Henann, D., Yang, J., Estrada, J., Bryngelson, S. H., Rodriguez Jr. M., (2025). "A theory and data integrated method for inertial microcavitation rheometry in soft materials". *The Third Joint SIAM/CAIMS Annual Meetings (AN25)*.

[T64] **Song, H.**, Subramaniam, A., Olson, B., Wu, A., Chandra, A., Bryngelson, S. H., Lele, S., (2025). "Parallel and GPU-optimized linear solver for compact difference schemes". *APS Global Physics Summit*.

[T63] **Wilfong, B.**, **Chu, T.**, McMullen, R., Koehler, T., Bryngelson, S. H., (2025). "Euler–Lagrange simulations of near-surface gas transport in vibrated bubbly flows". *Americal Physical Society, Division of Fluid Dynamics*.

[T62] Zhu, Z., Remillard, S., Abeid, B. A., Frolkin, D., Bryngelson, S. H., Yang, J., Rodriguez, M., Estrada, J. B., (2025). "Parsimonious inertial cavitation rheometry via bubble collapse time". *SES Annual Technical Meeting*.

[T61] Briney, S., Daoud, T., Bryngelson, S. H., Jackson, T. L., Balachandar, S., (2024). "Particle-resolved and Euler-Lagrange simulations of shock interaction with particle clusters using MFC". *Americal Physical Society, Division of Fluid Dynamics*.

[T60] Carcana Barbosa, M., Chreim, J. R., Tong, Z., Yang, J., Bryngelson, S. H., Henann, D., Colonius, T., Rodriguez, M., (2024). "Numerical simulations of inertial microcavitation near a gel-water interface with finite elasticity and phase change". *Americal Physical Society, Division of Fluid Dynamics*.

[T59] **Chu, T.**, Estrada, J., Bryngelson, S. H., (2024). "Bayesian optimal design accelerates discovery of material properties from bubble dynamics". *Americal Physical Society, Division of Fluid Dynamics*.

[T58] Jawetz, C., Bryngelson, S. H., Alexeev, A., (2024). "A quantum lattice Boltzmann algorithm for simulating heat transfer with phase change". *Americal Physical Society, Division of Fluid Dynamics*.

[T57] **Manasreh, S.**, Bryngelson, S. H., (2024). "Blood flow through a microaneurysm". *Art of HPC, SC '24 The International Conference on High Performance Computing, Network, Storage, and Analysis*.

[T56] **Radhakrishnan, A.**, **Le Berre, H.**, **Wilfong, B.**, Budiardja, R., Abbott, S., Bryngelson, S. H., (2024). "Compressible multi-species flow simulation on OLCF Frontier via OpenACC". *American Physical Society, March Meeting*.

[T55] Remillard, S., Zhu, Z., Abeid, B., Froklin, D., Bryngelson, S. H., Yang, J., Estrada, J. B., Rodriguez, M., (2024). "Soft material mechanical property determination using a modified Rayleigh collapse time". *Americal Physical Society, Division of Fluid Dynamics*.

[T54] Sanchez, V., Abeid, B., Yang, J., Estrada, J., Henann, D., Bryngelson, S. H., Rodriguez, M., (2024). "Bayesian constitutive model selection for inertial microcavitation rheometry". *Americal Physical Society, Division of Fluid Dynamics*.

[T53] **Song, Z.**, Gard, B., Bryngelson, S. H., (2024). "Incompressible flow simulation via a hybrid quantum-classical approach and variational algorithm". *American Physical Society, March Meeting*.

[T52] **Wilfong, B.**, **Chu, T.**, McMullen, R. M., Koehler, T., Bryngelson, S. H., (2024). "Hydrodynamic instability and breakup of a liquid-gas interface via vibration". *Americal Physical Society, Division of Fluid Dynamics*.

[T51] **Arias, J. E.**, Bryngelson, S. H., (2023). "Towards stable shock-capturing via radial basis finite differences on unstructured point clouds". *Americal Physical Society, Division of Fluid Dynamics*.

[T50] Bryngelson, S. H. (2023). "Stochastic sub-grid methods for multiphase flows at scale". *Workshop on Compressible Multiphase Flows, Stanford University*.

[T49] Bryngelson, S. H., **Le Berre, H.**, **Radhakrishnan, A.,** (2023). "Compressible multiphase flow simulation at near-exascale via a scalable GPU implementation". *American Physical Society, March Meeting*.

[T48] Bryngelson, S. H., Schäfer, F., Liu, J., Mani, A., (2023). "Super-spectral operator recovery via the fast macroscopic forcing method". *SIAM Computational Science and Engineering*.

[T47] Firouznia, M., Bryngelson, S. H., Saintillan, D., (2023). "A spectral boundary element method for interfacially driven flows". *8th Micro and Nano Flows Conference*.

[T46] **Kocherla, S.**, Bryngelson, S. H., (2023). "Reducing quantum resources for the quantum lattice Boltzmann method". *Americal Physical Society, Division of Fluid Dynamics*.

[T45] **Lee, H.**, Bryngelson, S. H., Colonius, T., (2023). "Cavitation inception in a turbulent mixing layer". *Americal Physical Society, Division of Fluid Dynamics*.

[T44] Liu, J., Schäfer, F., Bryngelson, S. H., Zaki, T., Mani, A., (2023). "Adjoint macroscopic forcing method for computing the nonlocal eddy viscosity in a turbulent channel flow". *Americal Physical Society, Division of Fluid Dynamics*.

[T43] Liu, J., Schäfer, F., Bryngelson, S. H., Zaki, T., Mani, A., (2023). "Targeted quantification of nonlocal closure operators using an adjoint-based macroscopic forcing method". *International Association for Computational Mechanics MMLDE-CSET*.

[T42] **Radhakrishnan, A.**, Bryngelson, S. H., (2023). "A statistical model for cavitation inception at the sub-grid scale". *Americal Physical Society, Division of Fluid Dynamics*.

[T41] Schäfer, F., Anandkumar, A., Bryngelson, S. H., **Kothari, Y.**, Owhadi, H., **Zeng, Q.**, Zheng, H., (2023). "Competitive gradient descent algorithms". *SIAM Computational Science and Engineering*.

[T40] **Song, Z.**, Bryngelson, S. H., (2023). "Tutorial: Solving partial differential equations (PDEs) with quantum computers". *QCE23: 2023 IEEE International Conference on Quantum Computing & Engineering*.

[T39] **Song, Z.**, Gard, B., Bryngelson, S. H., (2023). "Hybrid classical-quantum algorithm for solving the incompressible Navier–Stokes equations on quantum hardware". *Americal Physical Society, Division of Fluid Dynamics*.

[T38] **Arias, J. E.**, Bryngelson, S. H., (2022). "Radial-basis-function-based numerical methods for solving compressible flow equations at different Mach numbers". *American Physical Society, Division of Fluid Dynamics*.

[T37] **Bati, A.**, Bryngelson, S. H., (2022). "RoseNNa: A performant library for portable neural network inference with application to CFD". *American Physical Society, Division of Fluid Dynamics*.

[T36] Bryngelson, S. H. (2022). "Fast integration methods for averaging bubble dynamics at sub-grid scales". *19th U.S. National Congress on Theoretical and Applied Mechanics*.

[T35] Bryngelson, S. H., Schäfer, F., Liu, J., Mani, A., (2022). "Fast Macroscopic Forcing: Exploiting locality for operator recovery". *American Physical Society, Division of Fluid Dynamics*.

[T34] **Chrit, F. E.**, **Kocherla, S.**, Adams, A., Young, J., Alexeev, A., Bryngelson, S. H., (2022). "Quantum lattice algorithms for solving partial differential equations". *17th Conference on Theory of Quantum Computation, Communication, and Cryptography*.

[T33] **Chrit, F. E.**, **Kocherla, S.**, Alexeev, A., Bryngelson, S. H., (2022). "Quantum lattice gas algorithm for fluid flow simulations". *American Physical Society, Division of Fluid Dynamics*.

[T32] Colonius, T., Bryngelson, S. H., (2022). "Hybrid quadrature moment methods for polydisperse cavitating flows". *1st European–American–Japanese Two-Phase Flow Group Meeting*.

[T31] Firouznia, M., Bryngelson, S. H., Saintillan, D., (2022). "A spectral boundary integral method for simulating electrohydrodynamic flows in liquid droplets". *American Physical Society, Division of Fluid Dynamics*.

[T30] **Panchal, A.**, **Radhakrishnan, A.**, Bryngelson, S. H., Menon, S., (2022). "A numerical comparison of 5-, 6-, and 7-equation Baer-Nunziato-based diffuse interface methods". *American Physical Society, Division of Fluid Dynamics*.

[T29] **Radhakrishnan, A.**, **Le Berre, H.**, Bryngelson, S. H., (2022). "Towards exascale multiphase compressible flow simulation via scalable interface capturing-based solvers and GPU acceleration". *American Physical Society, Division of Fluid Dynamics*.

[T28] Rodriguez, M., Bryngelson, S. H., (2022). "Cavitation bubble growth near an elastic object". *American Physical Society, Division of Fluid Dynamics*.

[T27] Rodriguez, M., Bryngelson, S. H., Colonius, T., (2022). "Numerical simulations of cavitation near an elastic object". *ECCOMAS Congress*.

[T26] Rodriguez, M., Spratt, J.-S., Bryngelson, S. H., Colonius, T., (2022). "Numerical simulations of cavitation bubble growth and collapse near a viscoelastic object". *19th U.S. National Congress on Theoretical and Applied Mechanics*.

[T25] Spratt, J., Rodriguez, M., Bryngelson, S. H., Colonius, T., (2022). "Numerical simulations of ablation mechanisms during focused ultrasound therapies". *American Physical Society, Division of Fluid Dynamics*.

[T24] **Zeng, Q.**, Bryngelson, S. H., Schäfer, F., (2022). "Competitive physics informed networks". *ICLR workshop "Gamification and Multiagent Solutions"*.

[T23] Bryngelson, S. H., Charalampopoulos, A., Fox, R. O., Sapsis, T., Colonius, T., (2021). "Bypassing quadrature moment method instability via recurrent neural networks with application to cavitating bubble dispersions". *American Physical Society, Division of Fluid Dynamics*. URL: https://vimeo.com/650700675/06006b48de.

[T22] Bryngelson, S. H., Charalampopoulos, A., Sapsis, T., Colonius, T., (2021). "Machine learned model for non-Gaussian cavitation statistics". *International Association for Computational Mechanics MMLDT-CSET*.

[T21] Bryngelson, S. H., Colonius, T., (2021). "Statistical model for cavitating polydisperse bubble clouds". *Journal of the Acoustical Society of America*. URL: https://vimeo.com/640933361/4f9d1469ce.

[T20] Bryngelson, S. H., Colonius, T., (2021). "Sub-grid population balance model for cavitating flows". *14th Southern California Flow Physics Symposium*.

[T19] Bryngelson, S. H., Wang, Q., Cisneros-Garibay, E., Colonius, T., (2021). "GPU-accelerated quadrature moment methods". *SIAM Annual Meeting*.

[T18] Rodriguez, M., Bryngelson, S. H., Colonius, T., (2021). "Acoustically induced bubble growth with phase change". *14th Southern California Flow Physics Symposium*.

[T17] Rodriguez, M., Bryngelson, S. H., Colonius, T., (2021). "Vapor and gas bubble growth with phase transition near a wall". *American Physical Society, Division of Fluid Dynamics*.

[T16] Spratt, J.-S., Rodriguez, M., Bryngelson, S. H., Cao, S., Colonius, T., (2021). "High fidelity single framework simulations of acoustic wave–bubble cloud–elastic solid interactions". *American Physical Society, Division of Fluid Dynamics*.

[T15] Spratt, J.-S., Rodriguez, M., Bryngelson, S. H., Cao, S., Colonius, T., (2021). "Numerical Simulations of burst-wave lithotripsy in an Eulerian framework". *14th Southern California Flow Physics Symposium*.

[T14] Spratt, J.-S., Rodriguez, M., Bryngelson, S. H., Cao, S., Colonius, T., (2021). "Single-framework simulations of acoustic-wave–bubble cloud–stone interactions". *Journal of the Acoustical Society of America*.

[T13] Bryngelson, S. H., Fox, R., Colonius, T., (2020). "Conditioned quadrature moment methods for cavitating bubble dispersions". *American Physical Society, Division of Fluid Dynamics*. URL: https://vimeo.com/640933407/2830fcf3e0.

[T12] Rodriguez, M., Bryngelson, S. H., Colonius, T., (2020). "Cavitation bubble growth with phase transition near a rigid wall". *American Physical Society, Division of Fluid Dynamics*.

[T11] Spratt, J.-S., Rodriguez, M., Bryngelson, S. H., Colonius, T., (2020). "A fully Eulerian simulation framework for cavitating bubble-clouds near viscoelastic materials". *American Physical Society, Division of Fluid Dynamics*.

[T10] Bryngelson, S. H., Charalampopoulos, A., Sapsis, T. P., Colonius, T., (2019). "Neural-network-augmented Gaussian moment method for the statistics of cavitating bubble populations". *American Physical Society, Division of Fluid Dynamics*.

[T9] Bryngelson, S. H., Colonius, T., (2019). "Annular and spiral bubble nets: A simulation-focused analysis of humpback whale feeding strategies". *Journal of the Acoustical Society of America, 146(4) 2771*.

[T8] Bryngelson, S. H., Colonius, T., (2019). "Simulations and acoustics of humpback whale bubble-net feeding". *13th Southern California Flow Physics Symposium*.

[T7] Trummler, T., Schmidmayer, K., Bryngelson, S. H., Colonius, T., (2019). "Simulations of a collapsing gas bubble above a crevice". *13th Southern California Flow Physics Symposium*.

[T6] Bryngelson, S. H., Colonius, T., (2018). "Modeling approaches for bubbly, cavitating flows". *American Physical Society, Division of Fluid Dynamics*.

[T5] Bryngelson, S. H., Freund, J. B., (2017). "Floquet stability of tank-treading and tumbling capsules in viscous shear flow". *American Physical Society, Division of Fluid Dynamics*.

[T4] Bryngelson, S. H., Freund, J. B., (2017). "Global stability of fully coupled capsule flow systems". *SIAM Computational Science and Engineering*.

[T3] Bryngelson, S. H., Freund, J. B., (2017). "Stability of flowing red blood cell trains". *Blood Flow*.

[T2] Bryngelson, S. H., Freund, J. B., (2016). "Stability and transition to chaos of regular capsule trains". *American Physical Society, Division of Fluid Dynamics*.

[T1] Bryngelson, S. H., Freund, J. B., (2015). "Buckling and its effect on the confined flow of a capsule suspension". *American Physical Society, Division of Fluid Dynamics*.

# 12 Software

We develop and maintain MFC, an exascale multiphase and multiphysics fluid flow solver:

| **Name** (click for Github repo.) | **Description** |
|---|---|
| MFC | Exascale multiphase flow solver — 2025 Gordon Bell Prize Finalist \| 200T grid points on 43K+ GPUs |
| homebrew-mfc | Homebrew MFC tap |

More generally, our open source software is located at github.com/comp-physics, below is an auto-generated listing:

| **Name** (click for Github repo.) | **Description** |
|---|---|
| roseNNa | Put neural networks in your (GPU) PDE solver |
| group-docs | Group syllabus |
| Riemann35.jl | GPU-based 35-moment kinetic solver |
| dg-igr-mfem | IGR-DG mini-app |
| HyQMOM.jl | HyQMOM 3D |
| Carleman-QLBM | Carleman quantum LBM at scale |
| PyQBMMlib | PyQBMMlib is a Python extension of QBMMlib |
| RBF-NS | Radial basis functions incompressible Navier-Stokes |
| Quantum-HRF-Tomography | Linear-time reconstruction of pure quantum states |
| Quantum_Heat_LBM | Heat transfer via LBM on quantum devies |
| QLBM-frugal | A resource frugal quantum lattice Boltzmann method |
| RBC3D | 3D Spectral boundary integral solver for cell-scale blood flow |
| MeshfreeTrixi.jl | Meshfree extension to Trixi using RBF-based numerics |
| MeshfreeTrixiDev | Development and testing of MeshfreeTrixi.jl |
| NISQ-Quantum-CFD | CFD solve on a current quantum computer |
| Scientific-Visualization | Scientific visualization tutorials using Paraview |
| CPINN | Competitive Physics Informed Networks |
| fully-QLBM | Code accompanying quantum LBM paper |
| RadialBasisFiniteDifferences.jl | Library for efficient RBF-FD |
| 1d-shock-capturing-Euler | Simple shock capturing for 1D Euler eqns. |
| Quantum-PDE-Benchmark | Near-term quantum algorithm benchmarking for PDEs |
| deepOscillations | Flop-efficient neural integration |
| fast-mfm | Fast Macroscopic Forcing Method |
| awesome-numerics | Resources for learning about numerical methods. |
| qce23-qpde-tutorial | Solving PDEs with quantum algorithms: A tutorial at IEEE QCE 2023 |
| EnsAvg_1D_Tait | 1D Ensemble-averaging solver for dilute bubbly flows |
| awesome-modeling-simulation | Resources for learning about modeling and simulation |
| hip_stencil_code | Stencil code for AMD GPUs |

We also work on Inertial Microcaviation Rheometry (IMR) software:

| **Name** (click for Github repo.) | **Description** |
|---|---|
| IMR_nonspherical_dynamics | Perturbed cavitation bubble interfaces |
| Sliding_DA | Sliding-window data assimilation |
| IMRv2 | IMR version 2 |
| IMR_Bayesian_inference | IMR Bayesian Optimal Model Selection |
| IMR_Bayesian_design | IMR-based Bayesian Optimal Experimental Design |
| IMR_RBF_BOED | Accelerated BOED via RBFs |
| Intro-to-IMR | Simple codes to numerically simulate laser/ultrasound-induced inertial cavitation bubble dynamics in soft materials |
| IMR_data_assimilation | IMR with Data Assimilation |

| | |
|---|---|
| IMR_v1 | Vanilla IMR codebase |
| inca | InertialCav supported by J. Estrada's group |

Spencer's personal open source software is located at github.com/sbryngelson, below is an auto-generated listing:

| **Name** (click for Github repo.) | **Description** |
|---|---|
| ffmt | A fast Fortran formatter, written in Rust |
| academic-website-template | Static web template for academics/research groups |
| PyIMR | Differentiable IMR solvers + constitutive model suite |
| ane-guide | A reference manual for the Apple Neural Engine |
| ANEForge | Python runtime for the Apple Neural Engine |
| AGXForge | Native compiler and runtime for applegpu + tensor units |
| compiler-bugs | Some compiler issues I've found |
| microfd | A 3D compressible Navier-Stokes MPI/GPU solver in one short C file |
| verrou-dist | Build and distribute pinned Valgrind+Verrou |
| langchain-aneforge | LangChain embeddings for the ANE |
| whisper-aneforge | Whisper.cpp on the Apple Neural Engine |
| Fake-Beamer | You like Beamer? You need Powerpoint? Fake Beamer is for you! |
| spuma-cases | Scripts for SPUMA turbulence closures |
| spuma | Faster GPU OpenFOAM fork for RANS |
| figr | Finite volume IGR mini-app |
| georgia-tech-cv | LaTeX CV for Georgia Tech's RPT |
| georgia-tech-letterhead | Replicating the GT letterhead in LaTeX |
| elsevier-tikz-figures | Makes figures match Elsevier's style |